# NEIL SAWABUL JAZIL

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## OBJECTIVE

Vocational high school Software Engineering student focusing on robotics, embedded systems, and full-stack web development, with hands-on experience in Arduino, ESP32, React, Next.js, TypeScript, Laravel, and Supabase. Awarded 1st place in Robot Soccer at IARC ITS 2025. Seeking international university admission and scholarships to pursue Computer Science and contribute to automation technology in Indonesia.

## EDUCATION

**SMKN 2 Surabaya, Indonesia** 06/2024 – 05/2027

*Vocational High School – Software Engineering (RPL)*

- Majoring in Software Engineering; academic interests include computer programming, software development, mathematics, and technology.
- Vice Head of the Robotics Extracurricular Organization; active in robotics projects and competitions involving programming, embedded systems, and autonomous robots.

## PROJECT EXPERIENCE

**Line Follower Robot Project** 11/2025 – 05/2026

- Designed and built an autonomous line-following robot (microcontroller, infrared sensors, motors, motor drivers, power management); programmed and calibrated the control system for tracking accuracy, movement stability, and response speed through iterative testing and troubleshooting.

**Robotics & Embedded Systems Projects** 05/2025 – Present

- Research, design, and build robotic systems integrating microcontrollers, sensors, motors, motor drivers, and power-management components; program and test embedded systems on Arduino and ESP32, applying programming, electronics, mathematics, and engineering principles.

**Software Engineering & Web Development Projects** 01/2025 – Present

- Develop full-stack web applications (React, Next.js, TypeScript, Laravel, Supabase) with responsive interfaces, database integration, authentication, and role-based systems; solutions include point-of-sale (POS), e-commerce, and digital management systems.

**E-Commerce & Marketplace Web Project** 01/2025 – 03/2026

- Built a full-stack marketplace platform covering product management, transaction workflows, responsive UI, and backend services, with focus on usability, scalability, and performance.

## ORGANIZATIONAL & COMMUNITY ACTIVITIES

**ELITE (Electronic Innovation Center)** 07/2025 – Present

*Programmer & Electronics – robotics community for programming and electronics projects*

- Develop and program robotic systems (microcontrollers, sensors, motors, electronic components); test and troubleshoot hardware and software with team members to improve performance and reliability.

**Karang Taruna RT/RW** 08/2023 – Present

*PDD Staff (Publication, Documentation & Decoration) – neighbourhood youth organization*

- Create digital materials and document community activities and events through photos and videos; coordinate visual content and event materials to support promotion.

## AWARDS & HONORS

**16th Place – Transporter, Technocorner UGM** 03/2026

- Technology and robotics competition by Universitas Gadjah Mada; contributed to robot development, wiring, and testing with focus on electrical systems, microcontrollers, motor drivers, and programming.

**Participant – Line Follower, Microcontroller Automation Week 2025** 12/2025

- Built and tested an autonomous line-following robot (microcontrollers, sensors, motors, control algorithms) to improve accuracy and movement performance.

**5th Place – STEM Project Competition 2025** 11/2025

*Smart Metal Waste Sorter*

- Automated metal-waste detection and sorting system integrating hardware, software, and a web monitoring interface; presented as a practical waste-management solution applying electronics, programming, and automation.

**Participant – SoccerBot, Electro of Things Robocup 2025** 10/2025

- Contributed to electrical and programming systems (component integration, wiring, microcontroller programming, testing) of a soccer-playing robot.

**Participant – SoccerBot, BARONAS ITS 2025** 05/2025

- Robotics competition by Institut Teknologi Sepuluh Nopember; hands-on robotic system development, teamwork, and problem-solving in a competition-based engineering setting.

**1st Winner – Robot Soccer, IARC ITS 2025** 01/2025

- Indonesian Autonomous Robotics Competition by Institut Teknologi Sepuluh Nopember; main contributor for programming and electronic systems (control logic, microcontroller and component integration, movement and response testing), troubleshooting and optimizing performance with the team.

## TECHNICAL SKILLS

- Software: React, Next.js, TypeScript, Laravel, Supabase (responsive UI, databases, authentication, role-based systems).
- Hardware: Arduino, ESP32, sensors, motors, motor drivers, power management, control systems.

## LANGUAGES

- Indonesian: Native.
- English: Intermediate (IELTS preparation in progress).